A CONSTRUCTIBLE SHEAF ON THE PRO-ETALE SITE OF A NOETHERIAN SCHEME IS LISSE ON THE PIECES OF A FINITE PARTITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the construction of the associated graded sheaf over the Noetherian ring of quotients of powers of the ideal together with its surjectivity from degree zero, the partition making that graded sheaf locally constant, and the induction along the short exact sequences propagating local constancy back to the original sheaf.

Editorial note. The label lemma-Noetherian-constructible names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-Noetherian-constructible –start 5731 –end 5780 –local-id 3585 returned RESULT=OK (L3585 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Comparison of constructible sheaves on the pro-etale and etale sites. Same chapter, the lemma identifying constructible sheaves on the pro-etale site with the pullbacks of constructible etale sheaves. Supplies the existence of the constructible etale sheaves G_n whose pullbacks are the associated graded pieces I^n F / I^{n+1} F.
- Constructibility of a sheaf of modules over a Noetherian graded ring generated by its degree zero part. Etale Cohomology, the proposition that a sheaf of R-modules generated by the pullback of a constructible sheaf is constructible over a Noetherian scheme. Supplies the constructibility of the direct sum G of the G_n as a sheaf of R-modules.
- Existence of a partition on which a constructible sheaf of finite type is locally constant. Etale Cohomology, the lemma producing a finite partition into locally closed subschemes on which a constructible sheaf of finite type is locally constant. Supplies the partition by locally closed subschemes on which G becomes locally constant.
- Local constancy of the kernel of a surjection of locally constant sheaves of finite type. Modules on Sites, the lemma that the kernel of a surjection between locally constant sheaves of finite type is again locally constant

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of finite type. Supplies the induction step showing that the successive quotients I^n F / I^{n+1} F are locally constant, and hence that F is lisse.

Source reference. Original lemma 095E: sources/stacks-chow/tags.txt, source lines 11582–11582. Prerequisite pointer invoked at proof line(s) 5767; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 09BH: sources/stacks-chow/tags.txt, source lines 11795–11795. Prerequisite pointer invoked at proof line(s) 5763; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09AK: sources/stacks-proetale/pinned-original-source.tex, source lines 5456–5456. Prerequisite pointer invoked at proof line(s) 5748; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 093U: sources/stacks-chow/tags.txt, source lines 11527–11527. Prerequisite pointer invoked at proof line(s) 5778; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a Noetherian scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let $\mathcal{F}$ be a constructible Λ-sheaf on $X_{pro\text{-}étale}$. Then there exists a finite partition $X = \coprod X_i$ by locally closed subschemes such that the restriction $\mathcal{F}|_{X_i}$ is lisse.*

Proof. Let $R = \bigoplus I^n/I^{n+1}$. Observe that R is a Noetherian ring. Since each of the sheaves $\mathcal{F}/I^n\mathcal{F}$ is a constructible sheaf of $\Lambda/I^n\Lambda$-modules also $I^n\mathcal{F}/I^{n+1}\mathcal{F}$ is a constructible sheaf of Λ/I-modules and hence the pullback of a constructible sheaf $\mathcal{G}_n$ on $X_{étale}$ by Lemma 09AK. Set $\mathcal{G} = \bigoplus \mathcal{G}_n$. This is a sheaf of R-modules on $X_{étale}$ and the map

$$\mathcal{G}_0 \otimes_{\Lambda/I} \underline{R} \longrightarrow \mathcal{G}$$

is surjective because the maps

$$\mathcal{F}/I\mathcal{F} \otimes \underline{I^n/I^{n+1}} \to I^n\mathcal{F}/I^{n+1}\mathcal{F}$$

are surjective. Hence $\mathcal{G}$ is a constructible sheaf of R-modules by Étale Cohomology, Proposition 09BH. Choose a partition $X = \coprod X_i$ such that $\mathcal{G}|_{X_i}$ is a locally constant sheaf of R-modules of finite type (Étale Cohomology, Lemma 095E). We claim this is a partition as in the lemma. Namely, replacing X by X_i we may assume $\mathcal{G}$ is locally constant. It follows that each of the sheaves $I^n\mathcal{F}/I^{n+1}\mathcal{F}$ is locally constant. Using the short exact sequences

$$0 \to I^n\mathcal{F}/I^{n+1}\mathcal{F} \to \mathcal{F}/I^{n+1}\mathcal{F} \to \mathcal{F}/I^n\mathcal{F} \to 0$$

induction and Modules on Sites, Lemma 093U the lemma follows. □